An Efficient and Lightweight CNN Model With Soft Quantification for Ship Detection in SAR Images

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***Abstract*— Convolutional neural networks (CNNs) have been widely used for synthetic aperture radar (SAR) target detection. Typical methods based on CNN have obtained favorable detection accuracy at the cost of high model complexity, and thus are difficult to be directly applied to real-time satellites on board as well as maritime rescue. To deal with this problem, this article proposes an efficient and lightweight target detection network incorporating soft quantization. First, to compensate for the lack of accuracy caused by lightweight networks, a feature fusion module called split bidirectional feature pyramid network is proposed to alleviate the interference of complex background on SAR images. Meanwhile, to adapt the lightweight network and the feature fusion module, a linear transformation module is presented to enhance the linear representation of the model via learnable parameters. Eventually, to make the model size smaller, a soft quantization algorithm is proposed to reduce the accuracy degradation caused by quantization errors. We validate the robustness of the model in several publicly available datasets. Experimental results show that our model achieves 97.0% detection accuracy on SAR ship detection dataset, with a 0.9% accuracy improvement compared to mainstream methods using less than 15× the number of parameters and less than 6× the number of flops.**

***Index Terms*— Lightweight network, ship detection, soft quantization, synthetic aperture radar (SAR).**

I. Introduction

With the development of synthetic aperture radar (SAR) technology from single-polarization and single-mode in the past to high-resolution, multipolarization, and multimode imaging radar, these large amounts of new SAR data contain increasingly rich information which is not limited by light, climate, clouds and fog. These image data provide a direct means for target detection and monitoring of various ground activities.

In recent years, the research of marine target detection and surveillance using SAR images has received high attention in the field of remote sensing. Effective identification and detection of ship targets using SAR images can monitor maritime traffic and improve the early warning function of sea defense. Traditional methods, such as constant false positive rate [1]–[6] and maximum stable region method [7], usually use hand-designed features to detect ships after sealand segmentation. However, these methods cannot achieve satisfactory accuracy. In recent years, object detection methods based on convolutional neural network (CNN) [8]–[11] have been applied to ship detection to obtain higher detection accuracy. They are generally divided into two categories: one-stage methods [12]–[16] and two-stage methods [17]–[20]. The advantage of the two-stage approach is higher accuracy, but it also suffers from complex and time-consuming training issues. The advantage of the one-stage method is end-to-end detection, which is fast but slightly less accurate than the two-stage method. In addition, methods based on feature extraction [21]–[24] are gradually being widely used. These methods combine image features for feature extraction to achieve better accuracy, but the robustness of the algorithm also deteriorates. To sum up, these methods all use deep CNN to extract features, which leads to a complex network structure and a large amount of computation. They do not take into account the inability to meet the actual resource requirements for real-time ship detection, maritime rescue, and emergency military decision-making.

Therefore, designing an efficient and lightweight CNN detection model is an effective way to solve this problem. It aims to achieve better detection accuracy through an efficient network, using less memory and computing power. In the field of object detection, most of the methods are simply replacing the typical target detector with a lightweight backbone, such as the MobileNet series [25]–[27] and the ShuffleNet series [28], [29], which usually reduces the computational effort of the model by replacing ordinary convolution with more efficient convolution. However, there are some problems with using lightweight networks directly. It usually compresses the number of channels through 1×1 convolution between the basic modules and then expands the number of channels, which to some extent limits its feature extraction capability and leads to a certain degree of accuracy degradation. Moreover, hardware storage resources are often limited in practical applications. Lightweight network can only reduce the number of model parameters, but cannot meet the limited memory resources and model size.

To solve the above problems, this article proposes an efficient and lightweight CNN model for ship detection in images. After building an SAR target detection network based on the lightweight network, we present a split bidirectional feature pyramid network (S-BiFPN) module to enhance the

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feature by suppressing the background information and noise interference. In addition, to increase the quantization interval, we design a linear transformation (LTM) module to stretch the numerical representation range of the head of the neural network and reduce the gap between network modules. Finally, the 8 bits quantized model is obtained through soft quantization (Soft-Q), which can greatly reduce the memory resources occupied by the model. Experimental result of ship detection on SAR ship detection datasets (SSDDs) [30] show that we achieve the best performance with fewer parameters and flops, demonstrating the effectiveness of our model.

In summary, the main contributions of this work are summarized as follows.

1) We design an efficient feature fusion module called S-BiFPN module, which can effectively solve the background interference and improve the accuracy of the model by using only 1/15 of the parameters.
2) From a quantitative point of view, LTM module is proposed to adopt the lightweight network and the feature fusion module, which can improve the performance of the model without increasing the other parameters.
3) We propose Soft-Q by simulating quantization during the training process, which can not only reduce the impact of quantization errors on the detection process but also improve the performance of the model slightly.
4) Without bells and whistles, our model achieves detection accuracies of up to 97.0% on SSDD dataset, using 15× lower parameters and 6× lower flops than the baseline.

II. Related Work

A. Object Detection

As one of the most important and challenging branches in the field of computer vision, the task of target detection is to simultaneously identify and localize instances of a certain class of semantic objects. The current approaches of deep learning in the field of target detection are mainly divided into two categories: two-stage and single-stage.

1) Two-Stage Detection Methods: They are mainly to generate the target region of interest (ROI) first, and then do classification and regression on the ROI. Region-CNN (RCNN) [31] successfully applied CNN network for target detection for the first time, but at the same time, there are many areas that can be optimized. Fast R-CNN [32] proposed ROI Pooling Layer to extract the corresponding features of each ROI in the global features, which still exists the problem of long time of the search algorithm. Faster R-CNN [33] used region proposal network to generate detection regions and suppress the overlapping regions by a great suppression algorithm, which can improve the accuracy. Feature Pyramid Network (FPN) [34] introduced a multiscale feature fusion method to enrich the semantic information of features. Mask R-CNN [35] used ROI Align instead of ROI Pooling to improve the detection accuracy and added a mask branch to extract object masks. However, all these methods have a drawback of repeated computations. In summary, the two-stage approaches can achieve better accuracy but still suffer from the time-consuming problem.

2) Single-Stage Detection Methods: They mainly use end-to-end CNN to directly predict the class and location of the target. You Only Look Once (YOLO)v1 [36] used the entire image as input to the network and performed a direct regression. In order to improve the accuracy of localization, YOLOv2 [37] proposed a new training method-joint training algorithm. In pursuit of real-time detection, YOLOv3 [38] achieved higher accuracy and faster detection by changing the network structure and using multiscale prediction. YOLOv4 [39] obtained an efficient pipeline by exploring the performance of many general algorithms on the network. Single Shot MultiBox Detector (SSD) [40] used a small convolutional kernel to predict the class and boundary offset of a set of bounding boxes on the feature map to balance accuracy and speed. In addition, subject to residual connectivity, Deconvolutional Single Shot Detector (DSSD) [41] utilized a deconvolution module to improve the robustness of object detection. Recently, some algorithms that do not rely on a predefined frame have been proposed. For example, CornerNet [42] introduced the upper-left and lower-right corners of the regression, and CenterNet [43] predicted the centroid of each object directly. These detection structures rely on deep CNN for feature extraction, and the performance of the network model is greatly improved as the depth of the network increases. Therefore the number of parameters of these methods is difficult to meet the actual resource requirements.

B. Lightweight Network

Lightweight networks aim to further reduce the number and complexity of model parameters based on maintaining accuracy, and are gradually becoming a research hotspot in computer vision. Lightweight networks contain both the exploration of network structure and the application of compression techniques, such as knowledge distillation, pruning, and quantization.

Design of lightweight network structure mainly applied efficient convolution to reduce the computation of convolutional networks. For example, the MobileNet series [25]–[27] built the network through depthwise separable convolution and introduces residual connections into the model, which greatly reduces the amount of parameters of the model. Similarly, ShuffleNet series [28], [29], applied group convolution on channels to reduce the number of parameters and increased the information flow between grouped convolutions by shuffle operations. GhostNet [44] designed a staged convolutional computation module to reduce the computation by generating redundant feature maps through linear computation. Neural network structure design can greatly reduce the number of parameters of the network, but cannot solve the memory occupation problem of the model.

Model compression techniques aim to satisfy the need to restore the performance of the original network as much as possible while using a small number of computational

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resources. Among them, the main idea of parameter quantization is to reduce memory and computational consumption by reducing the number of bits of model parameters. For example, the binary weight network [45] was a binarization network algorithm that used $+1$ and -1 representations for weights, and the XNOR-Net [46] network operated on both weights and input pairs binarization, and they suffered from the problem that training does not converge easily. To solve the problem of accuracy loss, Post-Training Piecewise Linear Quantization (PWLQ) [47] proposed a segmented linear quantization algorithm for the weights of a specific distribution. In addition, knowledge distillation [48] reduced the complexity of the compressed network structure by guiding the training of small models with large models, so that the small models retain the effective information on the large models. Pruning technique [49] was based on the idea that low weights have low impact and removed the parameters close to zero in the convolutional network to achieve the reduction of the computational and parameters of model. Among these three approaches, pruning and knowledge distillation only reduce the model size to achieve low memory resource usage. However, quantization reduces the memory size of data from the storage perspective, which is often more effective in practical applications.

C. Light Ship Target Detection

In recent years, lightweight ship detection network has gradually started to receive more attention. Li *et al.* [17] designed a lightweight basis network by improving R-CNN through inception module and jump connection. Wang *et al.* [50] used a segmentation algorithm to out-process images based on a lightweight network and gradually eliminated false alarms by multilevel feature cascading. Zhou and Piao [51] used lightweight network to replace the backbone network of YOLOv4 and used separable convolution to replace the standard convolution, which can effectively reduce the number of parameters while obtaining higher accuracy, but the number of parameters still cannot meet the practical needs.

These methods simply apply the basic modules of lightweight to improve the backbone of the target detection model. There are two problems with this approach: 1) replacing the deep CNN network with a lightweight network will inevitably cause a significant decrease in detection accuracy and 2) the memory consumption resources of the actual model are not considered, which is the key to limit the practical application of the detection model.

To obtain a lightweight ship target detection model, we use a lightweight network for the above methods (e.g., He *et al.* [35]) and apply separable convolution instead of the standard convolution. Further, S-BiFPN as well as LTM modules, are proposed to lighten the multiscale feature extraction module and detection head, respectively, to reduce the number of parameters in the model. More, we propose a soft quantization strategy to reduce the memory footprint of the model to meet the practical requirements while other methods do not consider this issue.

III. Method

A. Overview of the Method

In order to obtain a lightweight ship target detection model, we first reduce the number of parameters of the model by lightweight the multiscale pyramid module and the detection head module, respectively, and then apply a soft quantization strategy to the whole model to further reduce the memory footprint of the model.

Fig. 1 illustrates the framework of the proposed model for ship detection in remote sensing images. Our model consists of three parts: 1) an S-BiFPN for feature fusion; 2) an LTM module to enhance the linear expression ability of the model; and 3) a Soft-Q module to reduce the memory space occupied by the model. This lightweight detection network consists of three parts: backbone, feature fusion, and head. The backbone is built on the classic lightweight detection network MobileNetv2 for feature extraction. Lightweight feature fusion module is added after MobileNetv2. It mainly contains two operations, one is a split operation, which is used to distinguish foreground and background information, and the other is a simplified BiFPN layer, which fuses multiscale feature semantic information and context information. The head part enhances the linear expression ability of the model and stretches the parameter range of the model through LTM module, and reduces the quantization error in the quantization process. Finally, we use the Soft-Q module to perform 8bit quantization on the model to reduce the memory space occupied by the model.

B. Split Bidirectional Feature Pyramid Module

The aim of multiscale feature fusion is to find a transformation f that aggregates semantic information of features at different scales. Our multiscale module consists of two main parts, a split operation and a BiFPN layer. The main function of split operation is for distinguishing background and foreground information. The simple BiFPN layer performs the fusion of semantic and contextual information.

1) Simple BiFPN Layer: The traditional feature pyramid network structure is naturally limited by the unidirectional information flow due to its top-down feature. To solve this problem, Path Aggregation Network (PANet) [52] adds an asymmetric bottom-up path, as shown in Fig. 2(b). PANet achieves better accuracy at the cost of more parameters and computations. To improve model efficiency, BiFPN [53] introduces cross-scale connectivity, repeating the same layer multiple times to achieve more advanced feature fusion. However, the recent You Only Look One-level Feature (YOLOF) [54] shows that the key to an effective multiscale fusion module is the idea of divide and conquer, not the fusion of information at different scales. Based on the consideration of structural lightweight, a simplified BiFPN layer is proposed in this article, as shown in Fig. 2(d). First, we use only the last three outputs (P5–P7) of the network for multiscale modular fusion. This is because our intuition is simple: if the network is efficient enough, the features in the current layer have already learned enough feature semantic information. The more layers are fused, the more semantic information and background interference information in different layers

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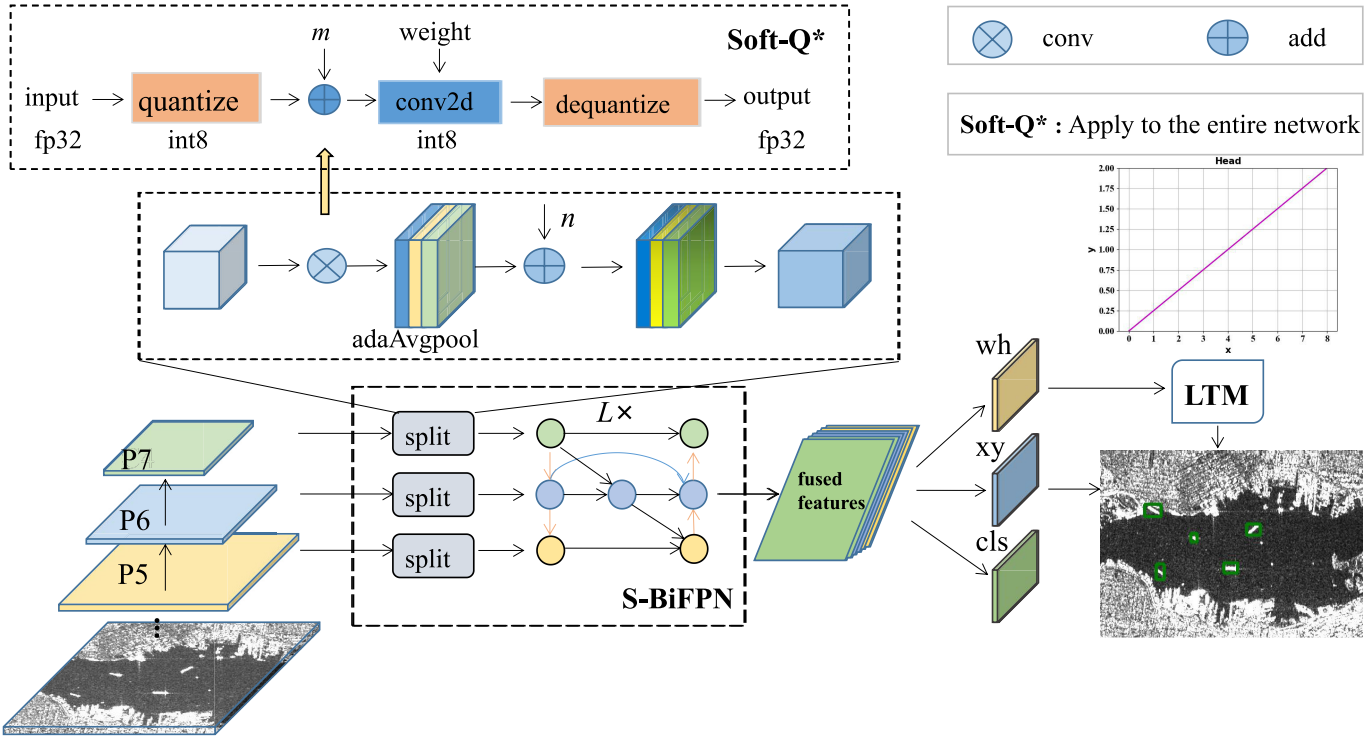


Fig. 1. Flowchart of the proposed method. First, feature maps of images are extracted by the backbone. Next, S-BiFPN module is applied to the feature map, which first removes the background and noise information by split operation, and then extracts the contextual information for feature fusion by using the feature pyramid structure connected via residuals. In addition, the linear representation of the model detection is enhanced by LTM module. Finally, soft quantization (Soft-Q) is applied to the whole model. Notes: for convenience, the figure only shows the process of applying soft-Q to a convolution, in fact, we apply soft-Q to all convolutional layers in the entire network.

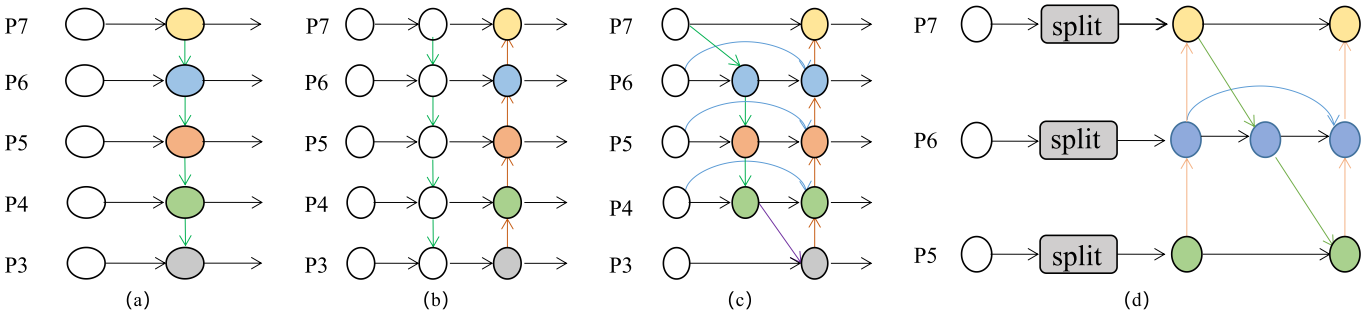


Fig. 2. Feature network design. (a) FPN [34] applies top-down paths to fuse multiscale features. (b) PANet [52] adds bottom-up paths on top of FPN. (c) BiFPN [53] introduces residual connections on top of PANet. (d) S-BiFPN adds split operation before simplifying BiFPN. Notes: P3-P7 represent the last five layers of output features of the model.

increase, which is not beneficial for later prediction. In addition, we still retain residual connections to learn contextual information by fusing features with other layers, while retaining the semantic information learned in the current layer.

2) Split Operation: As mentioned above, SAR images used for ship detection are subject to complex background interference, and the existing feature pyramid methods cannot achieve a better fusion of features. Based on this, combined with the feature pyramid structure, we introduce a split operation for SAR images with background interference. Inspired by the recent attention mechanism that has obtained good results in target detection tasks, we perform a channel-by-channel adaptive threshold splitting of the feature maps obtained via backbone to distinguish foreground information from background information, while using a learnable parameter C to suppress the background information. The specific one can be expressed by

$$\bar{y} = y + \alpha C \tag{1}$$

$$\alpha = \begin{cases} 0, & \text{if } y \geq \text{threshold}(y) \\ -1, & \text{if } y < \text{threshold}(y) \end{cases} \tag{2}$$

$$\text{threshold}(y) = \text{Gaussian}_{i*i}(y) \tag{3}$$

where y denotes the original pixel value of the channel feature map, $\bar{y}$ denotes the pixel value after split operation, threshold (x) represents the threshold function, and C denotes the learnable suppression parameter. Gaussian_{i*i} represents a Gaussian sum over the pixel values in the neighborhood $i * i$. Of course, you can also use mean instead.

Intuitively, as a basic module, S-BiFPN has several advantages: 1) combining the characteristics of SAR images

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themselves, the refinement of feature layers with fewer operations can efficiently obtain the semantic and contextual information of the target and achieve feature fusion and 2) using the learnable parameter C and the addition and subtraction operations instead of the complex convolution operations, which can greatly reduce the number of model operations.

C. LTM Module

In recent years, there have been many studies on target detection heads. Nowadays, common prediction method used is the detection head prediction of YOLOv3 [38] with the following expression:

$$b_x = \sigma(t_x) + c_x, \quad b_y = \sigma(t_y) + c_y \tag{4}$$

$$b_w = p_w * e^{t_w}, \quad b_h = p_h * e^{t_h}. \tag{5}$$

It predicts the absolute offset of the center point and the relative offset of the width and height. However, this method leaves the width and height unrestricted, which may lead to nonconvergence of the gradient. To address this problem, the Sigmoid function used in YOLOv5 fixes the error and additionally adds 0.5 to the centroid coordinate to keep the centroid position constant. In addition to the problem of conflict between classification and regression in the detection task, Exceeding You Only Look Once (YOLOX) [55] uses the decoupling head method, which can speed up the model convergence, but brings the additional number of parameters and operational complexity.

The above study shows that the detection head needs to be adapted to the lightweight network. Considering that an efficient lightweight neural network is an approximate linear representation of a deep CNN, we also make some improvements to the detection head. Specifically, we choose to use linear functions instead of nonlinear functions (e.g., Sigmoid function) to increase the network capacity of the lightweight network and compensate for the lack of spatial induction capacity, as shown in Fig. 3. Our linear detection head is represented as

$$b_w = p_w + t_w, \quad b_h = p_h + t_h. \tag{6}$$

Furthermore, inspired by the idea of knowledge distillation, we find the best linear expressivity of the model by increasing the variable parameter n. Thus, the final prediction expression can be expressed as

$$b_w = p_w + \frac{t_w}{n}, \quad b_h = p_h + \frac{t_h}{n}. \tag{7}$$

D. Soft Quantization

Neural network quantization aims to represent the parametric information of a neural network with fewer bits. It has been demonstrated that neural networks are over-parameterized such that the networks contain more redundant information, and thus cropping this redundant information does not lead to significant accuracy loss. The essence of quantization is the readjustment of the range of values. The basic steps of neural network quantization can be represented as: assuming that the input is R and the quantized data is Q, the floating point is quantized to a fixed point

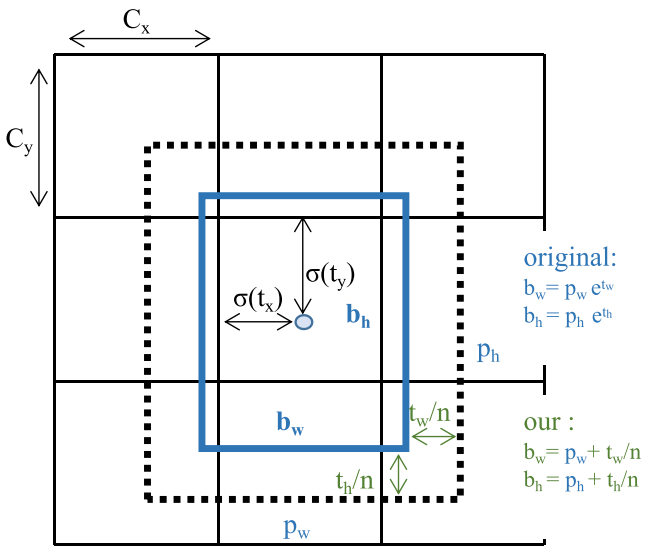


Fig. 3. Schematic of linear boundary prediction. We use the sigmoid function to predict the center coordinates of the box relative to the filter, and use a linear function to predict the offset of the width and height of the box. This figure is modified from YOLOv3 [38]. the original formula is the blue part and our modified one is the green part.

$$Q = \text{round}\left(\frac{R}{S} + Z\right) \tag{8}$$

where S is the minimum scale after fixed-point quantization and Z denotes the quantized fixed-point value corresponding to zero floating-point value. S and Z are calculated as follows:

$$S = \frac{R_{\max} - R_{\min}}{Q_{\max} - Q_{\min}} \tag{9}$$

$$Z = \text{round}\left(Q_{\max} - \frac{R_{\max}}{S}\right) \tag{10}$$

where $R_{\max}$ and $R_{\min}$ denote the upper and lower bounds of the floating point value. $Q_{\max}$ and $Q_{\min}$ represent the maximum and minimum values of the fixed-point values. Similarly, after compressing the transmission, the fixed-point can be inversely quantized into a floating-point representation as follows:

$$R = (Q - Z) * S. \tag{11}$$

The traditional quantization itself is bound to produce quantization error due to the presence of round operation, which causes the decrease of accuracy of the CNN. The most common round operation is round nearest to even (RNE). Although it is simple, it can lead to the problem of unstable training convergence. For this problem, stochastic rounding treats the value of each pixel as a probability to round. In particular, adaptive rounding analyzes the quantization error caused by rounding theoretically, obtains a learnable parameter V from the quantization loss, and quantizes the trained weights statically, which can reduce the accuracy loss. Obviously, they does not consider the impact of the quantization operation on the detection task.

We believe that the minimum quantization loss does not yield the best accuracy in the detection task. Specifically,

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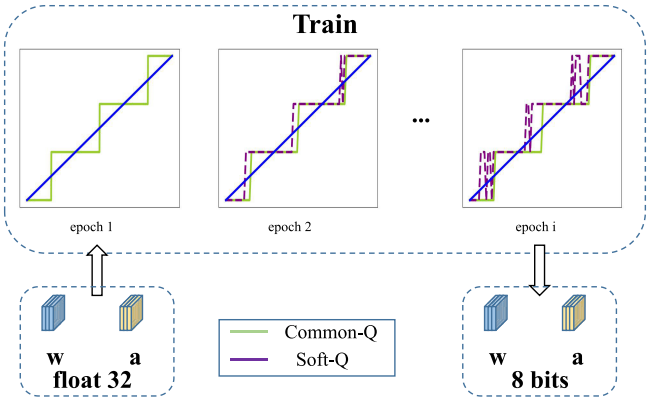


Fig. 4. Overview of soft quantization. First, we train a 32-bit floating-point model, and then apply soft quantization to simulate the quantization process and evolve it automatically in each epoch. In contrast to hard uniform quantization (green line), our soft quantization (purple line) allows for some worthwhile jumps in the quantization process.

as shown in Fig. 4, we simulate the quantization process during training and learn the changeable parameters to dynamically adjust the result of pixel value for each channel, which adjust feature map distribution is as similar as possible to the original feature map distribution. Our soft quantization formula is as follows:

$$Q = \text{round}\left(\frac{R}{S} + Z\right) + m \tag{12}$$

where m is a learnable parameter. Unlike adaptive rounding, we update the parameters by using the loss of the detection task rather than quantified loss.

IV. Experiments

A. Dataset

We generally test three datasets, including SSDD, high-resolution SAR ship objection detection dataset-2.0 (AIR-SARShip-2.0) [56], and SAR-Ship datasets [57]. These SAR images have a wide range of scenes, including harbors, islands, sea surfaces with different levels of sea state, etc. For SSDD dataset, which is the earliest publicly available dataset by time and includes 1160 images, we follow the same scheme as in Li *et al.* [30] to generate the training and test sets. For AIR-SARShip2.0 dataset, it adds more data onto version 1.0. The dataset contains 300 images (size 1000 × 1000) from the Gaofen-3 satellite. SAR-Ship dataset is composed of 43 819 images (size 256 × 256) from Gaofen-3 and Sentinel-1. For the latter two datasets, to be as consistent as possible with the division ratio of SSDD dataset, we both use a random selection of 80% of the whole dataset as the training set and 20% of the whole dataset as the test set.

B. Implementation Details

1) Technical Details: We use the PyTorch framework to implement the proposed method. To run the algorithm, we set up a deep learning environment for PyTorch on Ubuntu v16.04 system, i.e., Compute Unified Device Architecture (CUDA)v10.0 + cuDNN v7.0 + PyTorch. We apply MobiliNetV2 + PANet as our baseline. All experiments are performed on TITAN XP with 24 GB of memory. Pretrained MobileNetv2 network weights are used to initialize the backbone network. During training, the input image is resized to 416 × 416. The training model is trained using the Stochastic Gradient Descent (SGD) algorithm with a momentum of 0.9 and 0.0005 weight decay. The initialized learning rate is set to 1e-4, the minimum value of the learning rate is set to 1e-6, and the learning rate is decayed using the cosine annealing algorithm. The whole training process have 100 epochs with a batch size of 8. Incidentally, to evaluate the performance of different methods, the IoU value is constantly set to 0.5, which is used as the main evaluation metric [8].

TABLE I
Train Setting

config	setting
input image	416×416
init learning rate	1e-4
learning rate schedule	cosine decay
batch size	8
optimizer	SGD
optimizer momentum	0.9
weight decay	0.0005
training epochs	100
IoU	0.5

TABLE II
Experimental Results of Different L on SSDD Dataset

Parameters	mAP(%)
baseline	93.6
L=1	94.2
L=3	96.6
L=5	93.8
L=7	94.8

2) Parametric Analysis: The proposed method uses S-BiFPN module, where the parameter L affects the performance of the model. As shown in Table II, we used different L for several experiments. It can be observed that the model with different L achieves different levels of improvement compared to baseline (93.6%). The results show that the mAverage Precision (AP) on SSDD dataset first increases as L increases, reaches a peak and then gradually decreases. Therefore, we achieve the highest accuracy at $L = 3$. Intuitively, the simplified BiFPN output channel we used is 64.

C. Ablations Study

Our method consists of three parts, including a split bidirectional feature pyramid module, an LTM module, and soft quantization. To analyze the importance of each part, an ablation experiment is performed in Table III.

1) LTM Module: The linear conversion module increased the mAP by 1.4% (from 93.6% to 95.0%) compared to the baseline. This result indicates that enhancing the linear expression of the lightweight network can effectively compensate for its own weak spatial induction bias ability. In addition,

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TABLE III
EXPERIMENTAL RESULTS OF EACH COMPONENT ON SSDD DATASET

LTM	S-BiFPN	Soft-Q	mAP(%)	Params(M)	Model size
			93.6	46.26	185.05M
✓			95.0	46.26	185.05M
	✓		96.6	3.22	12.88M
✓	✓		96.8	3.22	12.88M
✓	✓	✓	97.0	3.22	3.22M

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT FEATURE FUSION MODULES ON SSDD DATASET

Fushion Module	mAP(%)	Params(M)	Flops(G)
PANet	93.6	46.26	8.71
FPN	94.8	7.63	4.14
BiFPN(L=3)	93.1	3.22	1.41
S-BiFPN (L=3)	96.6	3.22	1.41

adding S-BiFPN to the baseline is still able to improve the mAP by 0.2% (from 96.6% to 96.8%). The reason is that the LTM module is able to weaken the gap between the backbone module and the feature fusion module, which helps to improve the expressiveness of the model. More importantly, this method does not increase the model parameters, which meets the need of lightweight. This comparison experiment proves the importance of the LTM module.

2) Split Bidirectional Feature Pyramid Network: The S-BiFPN not only increases the mAP by 3% (from 93.6% to 96.6%) but also reduces 15× of parameters. This demonstrates that the module has the ability to learn semantic and contextual information about the target with fewer parameters and computational effort. With this lightweight feature fusion module, rich feature information can be learned and efficient feature fusion can be performed.

3) Soft Quantization: Soft quantization is able to improve the mAP by 0.2% (from 96.8% to 97.0%) and use less memory size to store the model weights. This shows that the method can effectively represent the model parameters in 8 bits, thus reducing the memory size occupied to meet specific resource requirements. In conclusion, this method can effectively reduce the memory footprint of the model without affecting or even improving the mAP of the detection model.

4) Compare With Other Methods: As shown in Table IV, we conduct the experiments by replacing the feature fusion module. To ensure the fairness of the experiment, we set output channels of the feature fusion module to 64 and fix parameter $L = 3$ uniformly. Experimental results show that, on the one hand, our S-BiFPN obtains the best mAP 96.6%, which is higher than FPN 94.8% (+1.8%), PANet 93.6% (+3%), and BiFPN 93.1% (+3.5%). On the other hand, our S-BiFPN also uses fewer parameters and fewer flops compared to other feature fusion modules. This shows that our S-BiFPN is lightweight and effective.

In order to verify the effectiveness of our Soft-Q, we do some comparative experiments with advanced methods.

TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT QUANTIZATION METHODS APPLIED TO THE NETWORK (MOBILENETV2 + S-BIFPN + LTM) ON SSDD DATASET

Quantization Method	mAP(%)	Size (M)
Without quantization	96.8	12.88
PACT	95.4	3.22
PWLQ	96.1	3.22
Bias absorption	96.4	3.22
Soft-Q	97.0	3.22

TABLE VI
EXPERIMENTAL RESULTS OF THE HEAD BASED ON NETWORK (MOBILENETV2 + S-BIFPN) ON SSDD DATASET

Head	mAP(%)	Size (M)	Flops(G)
Yolov3	96.6	3.22	1.41
Sqrt-head	92.4	3.22	1.41
Yolov5	95.9	3.22	1.41
LTM	96.8	3.22	1.41

As shown in Table V, we perform 8-bit quantization experiments on the network after adding LTM and S-BiFPN. Experimental results show that Parameterized Clipping Activation (PACT) (linear quantization), PWLQ (nonlinear quantization), and Bias absorption (data-free quantization) all yield results lower than the mAP of the original network (96.8%) on SSDD dataset, while the mAP is improved by 0.2% with our method, which proves the effectiveness of soft quantification.

To verify the effectiveness of the LTM module, we have done some comparison experiments. As shown in Table VI, compared with other heads, LTM can obtain the best accuracy without adding extra computation.

D. Comparison With the State-of-the-Art Methods

To make a fair comparison with the corresponding methods, we test them under the same conditions on three datasets, i.e., SSDD, AIR-SARShip2.0 and SAR-Ship dataset. As shown in Table VII, compared with the baseline, on the one hand, our model achieves 97.0% (+3.4%), 77.0% (+26.3%), and 95.7% (+1.3%) in the three datasets. On the other hand, the number of parameters of our model is reduced by 15×, the flops is reduced by 6×, and the running time is reduced by 4×. From the overall comparison, our method obtains stable performance with fewer parameters and can meet the requirements of real-time detection.

1) Compared to Mainstream Detectors: Our method is lower than 82.1% of YOLOv4 on the AIR-SARShip2.0 dataset, while the best accuracy is achieved on SSDD dataset and SAR-Ship dataset. Specifically, for the sake of lightweight, the number of parameters and flops of our model are reduced by an order of magnitude compared with the mainstream methods, which indicates that our method can effectively reduce the number of parameters and flops of the model.

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TABLE VII
COMPARISONS WITH STATE-OF-THE-ART METHODS ON SAR IMAGES

Method	SSDD			AIR-SARShip2.0			SAR-Ship			Params(M)	Flops(G)	Time(ms)
	mAP	Rec	Pre	mAP	Rec	Pre	mAP	Rec	Pre			
SSD512 [40]	94.0	94.4	82.7	75.7	76.8	64.9	94.2	93.1	80.2	24.2	87.7	30.2
YOLOv3 [38]	95.0	95.3	83.2	68.0	69.9	60.4	94.4	93.3	78.9	61.5	49.6	10.4
YOLOv4 [39]	96.1	96.4	84.3	**82.1**	**83.6**	**68.3**	93.2	92.5	81.6	64.3	45.3	12.9
CenterNet++ [13]	95.1	95.2	83.3	73.9	76.4	64.0	94.9	93.8	81.2	20.3	26.6	21.5
Libra R-CNN [20]	90.4	92.4	79.8	72.5	76.1	65.5	87.7	90.9	78.3	41.4	46.7	57.2
Faster R-CNN [33]	90.6	92.6	80.2	75.7	76.2	65.3	91.0	92.4	77.2	60.1	31.1	25.1
Cascade R-CNN [58]	90.4	93.1	79.3	74.7	77.5	64.4	92.0	92.5	80.1	87.9	58.9	34.0
MobileNetv2 [26]	94.7	93.3	78.7	61.2	66.7	45.6	92.5	91.2	80.9	9.66	2.39	7.5
MobileNetv3 [27]	89.8	86.6	73.6	51.3	51.8	56.8	90.0	87.7	73.0	5.71	1.03	7.4
ShuffleNetv1 [28]	94.2	93.1	80.7	60.2	61.1	52.8	92.9	92.4	77.4	7.34	2.62	**7.2**
ShuffleNetv2 [29]	94.8	94.8	79.3	53.2	56.9	49.6	92.8	92.2	81.2	6.29	1.64	7.2
GhostNet [44]	92.1	92.9	83.9	49.0	47.6	57.3	91.7	90.9	82.0	3.61	**0.56**	9.1
L-YOLOv4 [51]	95.3	93.9	88.3	68.7	69.3	62.0	95.5	92.6	81.7	43.0	9.3	42.1
Lira-YOLO [14]	86.3	87.3	70.6	47.8	52.5	47.5	90.9	89.0	83.1	4.3	2.98	13.7
baseline	93.6	91.8	86.2	50.7	51.6	49.4	94.4	93.6	79.7	46.3	8.7	34.1
ours	**97.0**	**97.4**	**89.1**	77.0	81.2	63.9	**95.7**	**95.3**	**88.7**	**3.22**	1.4	9.2

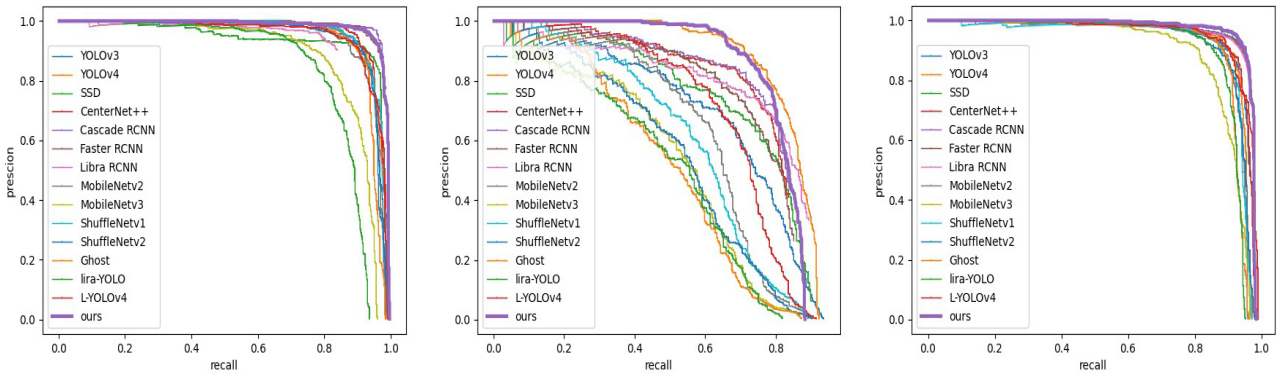


Fig. 5. P-R curves of different models on three datasets.

On the SSDD dataset, our method achieves better results compared to baseline, i.e., 97.0% mAP (+3.4%), 97.5% recall (+5.7%), and 89.1% precision (+2.9%). Specifically, the S-BiFPN module distinguishes foreground and background information by adaptive thresholding and suppresses the background information to highlight the foreground information, which is more conducive to the extraction of foreground information, which is very useful for ship target detection. Similarly, the mAP of our method is generally higher than other mainstream detectors, i.e., the mAP of our model is higher than YOLOv4 by 0.9% yet is similar to Faster RCNN (+6.4%). Moreover, the recall and precision of our model reach 97.5% and 89.1%, respectively, which are significantly higher than 96.4% and 84.3% of YOLOv4 as well as 93.1% and 79.3% of the two-stage algorithm Cascade RCNN. The main reason is that these algorithms cannot distinguish foreground and background information well.

On the AIRSARShip2.0 dataset, our method achieves 77.0% mAP, 81.2% recall, and 63.9% precision, which is lower than 82.1% mAP (−5.1%), 83.6% recall (−2.4%), and 68.3% precision (−4.4%) of YOLOv4. Compared to our model (Params 3.22 M), YOLOv4 (Params 64.3 M) uses a heavy backbone and PANet to extract features, which goes against our original intention of designing a lightweight model and reminds us that a strong backbone and efficient multiscale networks are the keys to ship detection. However, it is higher than other methods to varying degrees, i.e., Faster RCNN achieves 75.7% mAP (+2.3%), and 76.2% recall (+5.0%).

In the experiments on the SAR-Ship dataset, our method achieves 95.7% mAP, 95.3% recall, and 88.7% precision, which is better than the Cascade R-CNN 92.0% mAP (+3.7%), 92.5% recall (+2.8%), and 80.1% precision (+8.6%). The recall of the single-stage algorithm is significantly higher than that of the two-stage algorithm, which

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implies the importance of multiscale feature extraction models in ship detection. Likewise, the recall of our model is 2.7% higher than the baseline. This is attributed to the suppression of background information by adaptive thresholding in S-BiFPN.

Also, our model parameters are about 1/20 and 1/13 of that of YOLOv4 and Libra RCNN. This is because we use mobileNetv2 as backbone. In addition, the amount of our model parameters, flops is about 1/20 and 1/30 of that of YOLOv4. And the time is also lower than YOLOv4, which is that we use mobileNetv2 as backbone and apply depthwise separable convolution in the whole model.

2) Compared With the Lightweight Network: In all three datasets, the mAP, recall and precision of our method are higher than other lightweight networks using the least amount of parameters.

On the one hand, on the SSDD dataset, our model is higher than ShuffleNetv1 in mAP (+2.1%), recall (+2.7%), and precision (+9.8%), respectively. The mAP of 61.2% obtained by MobileNetv2 on the AIR-SARShip2.0 dataset is 15.8% lower than our result. Similarly, the recall of 66.7% and the precision of 45.6% are lower than our results of 14.5% and 18.3%. ShufflfleNetv1 on the SAR- Ship dataset achieves 92.9% mAP, 92.4% recall, and 77.4% precision, which is also lower than our method, respectively, by 2.8%, 2.9%, and 11.3%. The main reason for this is the addition of the S-BiFPN module to our lightweight model, which can extract the contextual information of features through multiscale connections. In addition, LTM also improves the linear representation ability of our model, which is beneficial to enhancing the regression of boundary features. These are very important for ship detection.

On the other hand, we also obtain a performance comparable to that of the lightweight network in terms of the number of parameters as well as flops. Specifically, our model uses only 3.22 million model parameters, which is the least compared to the lightweight network. In addition, our model requires 1.4 G flops, which is lower than GhostNet's 0.56 G and MobileNetv3's 1.03 G, but higher than ShuffleNetv1, ShuffleNetv2, and MobileNetv2. The main reason is that GhostNet uses feature multiplexing and only uses part of the channel for convolution, which reduces the amount of computation, while MobileNetv3 has a low amount of computation because the network search method is used to obtain a more compact network structure. The comparison results show that our model can meet the requirements of lightweight and requirements for detection accuracy.

3) Compared to the Lightweight Ship Target Detection Network: Compared with lightweight ship target detection network, our method achieves the best results in all measured metrics. First, the mAP of L-YOLOv4 is significantly higher than that of Lira-YOLO on all three datasets, which is because the coordinate attention used in L-YOLOv4 enhances the feature representation capability of the model. However, the mAP, recall, and precision of our model are higher than those of L-YOLOv4, which implies that our method can extract powerful feature representations for ship detection under complex backgrounds, i.e., the mAP recall and precision of our model are higher than L-YOLO by 1.7%, 4.4% and 0.8%, respectively, on the SSDD dataset. This shows that S-BiFPN can distinguish ship target features and background features, and suppress background features, thereby improving the recall and precision of ships.

Moreover, lira-YOLO is significantly lower than L-YOLO in parameters, flops and time, which uses a lightweight representation of the multiscale extraction network. Our model parameters, flops, and time are reduced by 25%, 50%, and 20% compared to lira-YOLO, respectively. The reason is that S-BiFPN is a more efficient and lightweight multiscale module.

Additionally, we visualize the precision-recall (P-R) curves of several different models (IoU threshold = 0.5). From left to right, the P-R curves of different models for SSDD, AIR-SARShip2.0, and SAR-Ship datasets are shown. It is well known that the model performs better when the model is closer to the upper right corner. The P-R curves are consistent with the experimental results, which validate the effectiveness of our method. In addition, when the recall is higher than 0.8, the precision of Cascade RCNN on the SSDD dataset and SAR-Ship dataset is better than the proposed model, but at the same time, the recall of Cascade RCNN decreases significantly, thus the final performance is lower than the proposed model, which is mentioned in this article of Cascade RCNN.

Finally, there are some analysis of the low results for the AIR-SARShip2.0 dataset. First of all, after analyzing the three datasets themselves, we found that the images in the AIR-SARShip2.0 dataset are larger, the average number of objects is larger, and the size of the real box varies more, which leads to more difficult samples and reduce the model accuracy to a certain extent.

However, YOLOv4 is still able to obtain high accuracy on the AIR-SARShip2.0 dataset. The reasons are mainly due to the following two reasons: the first is powerful feature extraction capability of the Cross Stage Partial Darknet53 (CSPDarknet53) network itself, and the second is the use of multiscale feature maps for detection. First, they can obtain much higher accuracy than that of the lightweight network regardless of the one-stage or two-stage methods, which fully demonstrates that a strong backbone is crucial to the accuracy of detection. Second, among the general detection methods, both YOLO series (YOLOV)v4 and SSD512 using multiscale features obtain better accuracy than the RCNN family of algorithms, which indicates that multiscale feature maps are also crucial to the accuracy of detection. YOLOV4 combines both of them, and the model itself is more optimized by various strategies so that it can obtain much higher accuracy than other models.

E. Visual Results

To demonstrate the advantages of our method over previous methods, we show some visual comparison results. As shown in Fig. 6, the results on simple background, complex background and dense target detection are shown from left to right. The first to third rows show the detection results of SSDD dataset, AIR-SARShip2.0 dataset, and SAR-Ship

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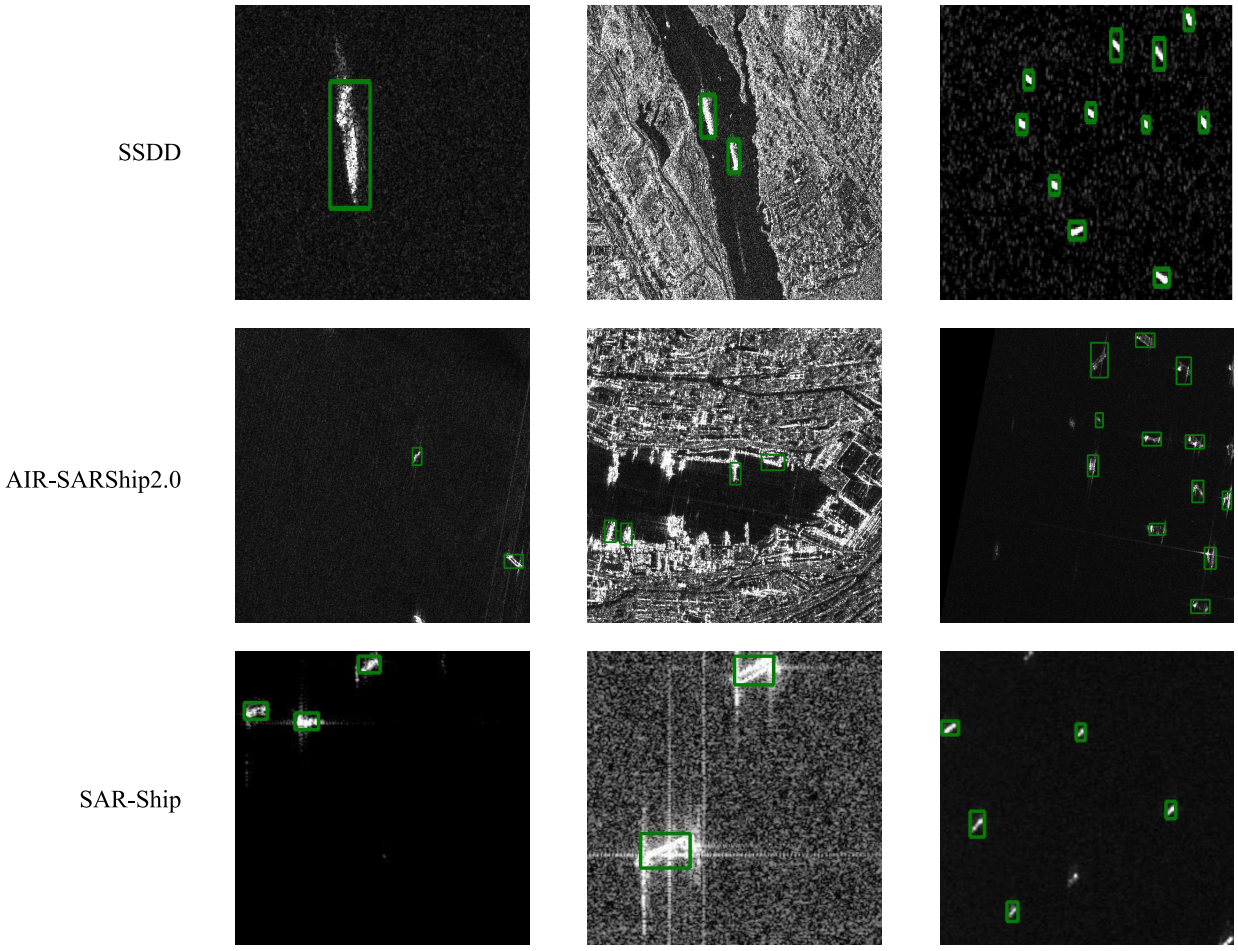


Fig. 6. Visualization of our method on three datasets. The first to third rows show the detection results of SSDD dataset, AIR-SARShip2.0 dataset, and SAR-Ship dataset under different scenarios, respectively.

dataset, respectively. In addition, it can be seen that using our method, we can obtain excellent detection results on all three datasets, which shows the robustness and effectiveness of our method on SAR datasets.

Fig. 7 shows some sample images of the detection results of different methods in different scenarios in the SSDD dataset. The first to third rows show the detection results of YOLOv4, Faster RCNN and our method, respectively. From left to right, the visualization results of simple background, complex background and dense target detection are shown, respectively. In comparison, YOLOv4 and Faster RCNN have different degrees of missed detection in all three scenarios, while our method is better at detecting all targets. The above result analysis shows that our method has better detection results in different scenes compared with other methods.

F. Analysis of Undesirable Results

We show some undesirable detection results in Fig. 8. In Fig. 8(a), some false positives are generated in the case of similar background noise to the sample. In Fig. 8(b), some missed detection results appear in the case of multi and small targets. This phenomenon is mainly due to the weak spatial sensing bias capability of the lightweight network itself, which is not friendly enough for small target detection.

TABLE VIII
COMPARISON RESULTS FOR DATASET

	SSDD	**SAR-Ship**	**AIR-SARship2.0**
Number of images	1160	43819	300
Number of targets	2540	59535	2040
ANTI	2.2	1.36	6.8
Image size	about 500×500	256×256	1000×1000

ANTI represents the average number of objects in a single image

In Fig. 8(c), two boxes are detected for the same target, mainly because the background is close to the target and has some similarity leading to duplicate detection. In addition, as shown in Table VII, our AIR-SARShip2.0 results are lower than YOLOv4, and lower than GhostNet and MobileNetv3 in terms of flops. The above results show that our model still has some room for improvement in terms of accuracy and lightness. On the one hand, for the problem of accuracy, we believe that it can be solved by using networks with added attention modules, such as building networks based on transformer and introducing attention modules in the network. On the other hand, we will try to generate redundant feature maps using linear computation instead of convolution, as well as a channel

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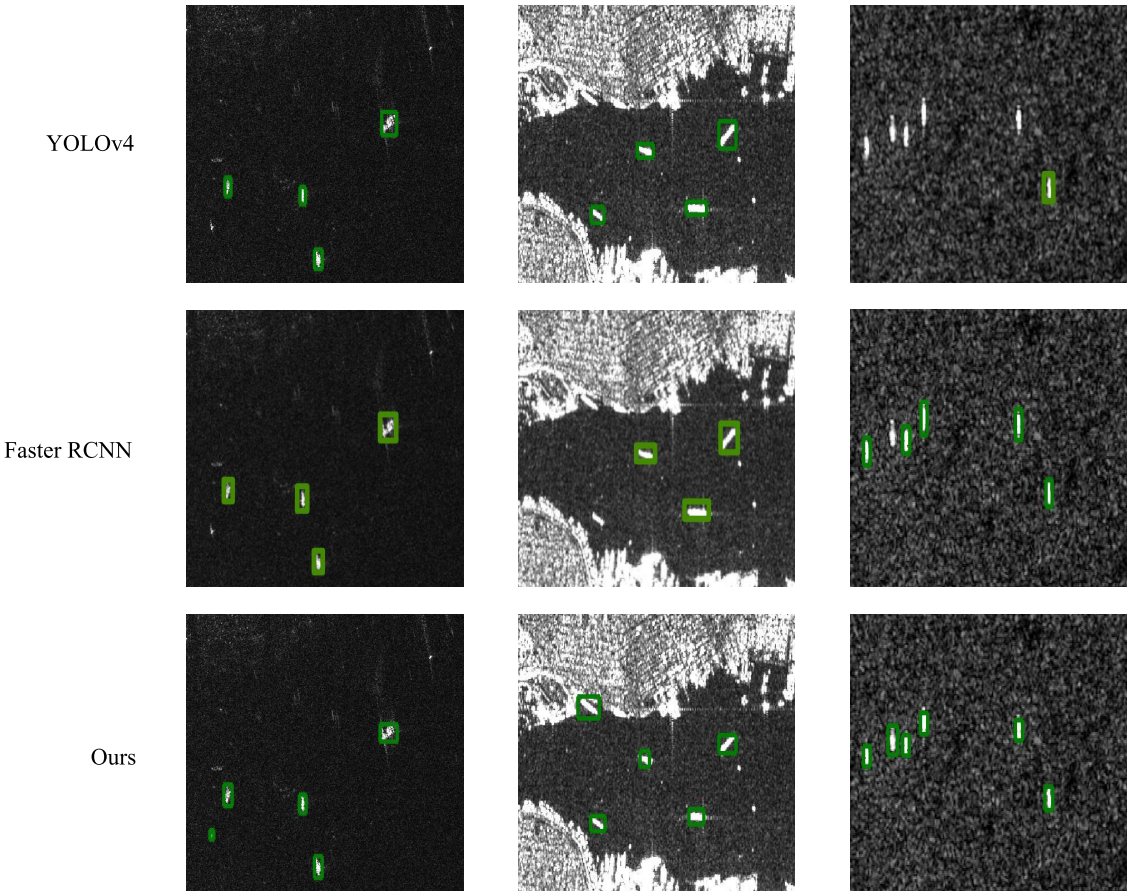


Fig. 7. Comparison of the detection results of different methods on SSDD dataset. From top to bottom are the test results of YOLOv4, Faster RCNN and our model, respectively.

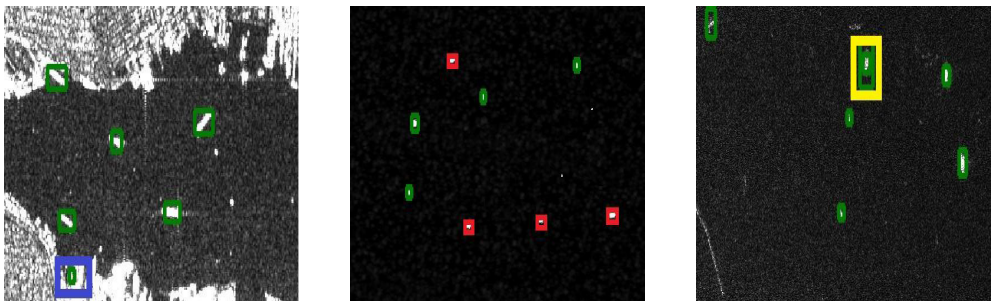


Fig. 8. Undesirable results of our method. The red box indicates the example of True Negatives (FN). The blue boxes indicate the results of false detection, and the green boxes indicate True Positives (TP). (a) Sample of FN. (b) Missed detection under multi small target. (c) Sample of repeat detections.

pruning approach to the network to solve the problem of flops. We will further investigate these in our future work.

V. Conclusion

In this article, we build a lightweight target detection network for ship detection in SAR images, and use quantization techniques to reduce the number of parameters and computational effort of the model. S-BiFPN is introduced to improve the detection accuracy by suppressing background and noise information. In addition, we propose an LTM module to improve the expressiveness of the detection head. Finally, we adopt soft quantization to reduce the impact of quantization errors on target detection. Experimental results show that our method achieves a performance comparable to state-of-the-art detectors using fewer parameters.

In future work, we will try to apply transformers to extract more efficient features. Moreover, we will also try other neural network compression techniques such as pruning and knowledge refinement techniques.

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